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Large Language Models, Small Labor Market Effects

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Abstract

We examine the early labor market impacts of AI chatbots by linking large-scale, representative adoption surveys to administrative labor records in Denmark. Using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after. These null results hold for intensive users, early adopters, workplaces with substantial investments, workers reporting large gains, flexible-pay occupations, and early-career jobs. Adoption is linked to occupational switching and task restructuring, but without net changes in hours or earnings. Our findings challenge narratives of imminent disruption from Generative AI.

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The emergence of AI chatbots marks the rise of Generative Artificial Intelligence (AI). By some measures, these technologies are already living up to the immense hype: AI chatbots have seen the fastest worker take-up of any new technology (Bick, Blandin and Deming, 2025; Humlum and Vestergaard, 2025), randomized controlled trials (RCTs) demonstrate substantial productivity gains for users (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023), and case studies indicate notable effects on online labor market platforms (Teutloff et al., 2025).¹

Yet even as Generative AI tools have become widespread, substantial disagreement remains about their labor market effects, with forecasts ranging from dramatic disruption within a few years (Axios, 2025) to negligible impacts over a decade (Acemoglu, 2025).²

This paper provides large-scale, representative evidence on the early labor market impacts of AI chatbots. In collaboration with Statistics Denmark, we conducted a series of extensive surveys of the adoption of AI chatbots in 11 highly exposed occupations, linking responses to administrative labor market records.³ Our latest survey round, conducted in late 2024, includes responses from 25,000 workers across 7,000 workplaces. Leveraging this unique data link, we provide the first comprehensive evidence on how AI chatbot adoption has affected labor market outcomes at both the worker and workplace levels.

To set the stage, we first describe the landscape of AI chatbot adoption. While chatbots are widespread, many otherwise similar workplaces and workers have taken markedly different paths: some employers still do not encourage use; others promote it with guidelines, enterprise tools, and training. Likewise, even among otherwise similar workers under identical policies, usage varies from none to daily. This rich variation—across both workplaces and workers—offers an opportunity to study how chatbot adoption relates

¹These effects on productivity and platform demand are remarkable in both magnitude—ranging from 15% to 50%—and speed, materializing within a few months.

²Anthropic CEO Dario Amodei recently predicted that AI could eliminate half of all entry-level white-collar jobs, driving unemployment to 10-20% within the next one to five years (Axios, 2025). In contrast, Acemoglu (2025) predicts that AI will increase annual TFP growth by less than 0.07 percentage points over the next ten years.

³Our list of occupations is accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketing professionals, office clerks, software developers, and teachers.

to labor market outcomes.

To examine these outcomes, we link our surveys to administrative records on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We begin by analyzing workers who use AI chatbots. If these tools enhance individual productivity, a natural question is whether such gains translate into higher earnings. Our first key finding is that AI chatbots have had minimal overall impact on workers who use them. Difference-in-differences estimates for earnings, recorded hours, and wages are all precisely estimated zeros, with confidence intervals ruling out average effects larger than 2%. These null results hold even for workers who use the tools daily, report substantial productivity gains, perform new AI-related tasks, and work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users, we find a clear link to occupational mobility: adopters are more likely to have transitioned into new occupations after the arrival of AI chatbots. This aligns with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023), and supports the view that Generative AI helps workers extend into new occupations (Autor, 2024). However, the effects are modest—amounting to 4% of full-time equivalent employment—and do not result in net gains in hours or earnings.

Next, we zoom out to the workplace level and examine whether employers that encourage and invest in AI chatbots have experienced differential changes in labor demand. If chatbots substitute for labor, they could lead to slower hiring or trigger layoffs; if they enhance productivity, firms might expand hiring. Yet we find no such differences: workplaces that encourage chatbot use show no differential changes in employment or